

Falsifiability



The method science uses to establish theories is based on induction – inferring a general rule from individual instances. Even after Hume pointed that induction cannot logically be used to show anything with certainty (see [The problem of induction](#)), scientists continued to use the methods of observation and experiment with remarkable success. But the problem of induction still nagged philosophers until the 1930s, when Karl Popper proposed a different way of viewing scientific methodology. He argued that while vast numbers of positive instances of an event cannot prove a scientific theory conclusively, a single negative instance can decisively

show it to be false. As an example, he gave the hypothesis: 'All swans are white'; this cannot be proved true by reference to any number of observations of white swans, but can be proved false or 'falsified' by the appearance of just one black swan. The true criterion of a scientific theory is not that it is inferred by induction, but that it should be falsifiable, capable of being shown as false by observation or experiment.